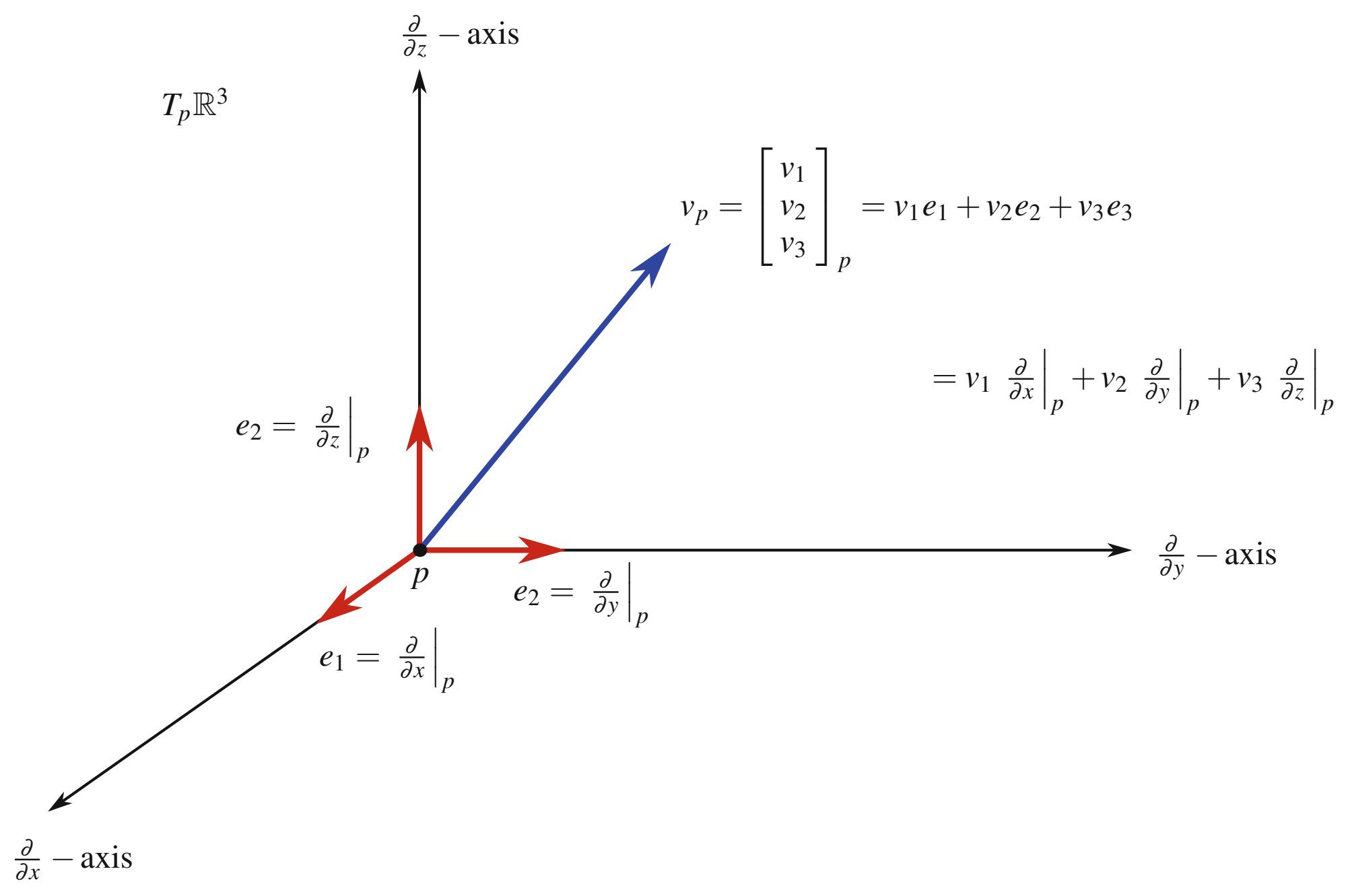

$\frac{\partial}{\partial z}$ – axis
$T_p\mathbb{R}^3$
$v_p = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}_p = v_1 e_1 + v_2 e_2 + v_3 e_3$
$= v_1 \left.\frac{\partial}{\partial x}\right|_p + v_2 \left.\frac{\partial}{\partial y}\right|_p + v_3 \left.\frac{\partial}{\partial z}\right|_p$
$e_2 = \left.\frac{\partial}{\partial z}\right|_p$
p
$\frac{\partial}{\partial y}$ – axis
$e_2 = \left.\frac{\partial}{\partial y}\right|_p$
$e_1 = \left.\frac{\partial}{\partial x}\right|_p$
$\frac{\partial}{\partial x}$ – axis